

Assignment:-9

1. Write a C++ program to write number 1 to 200 in a data file NUM.TXT.

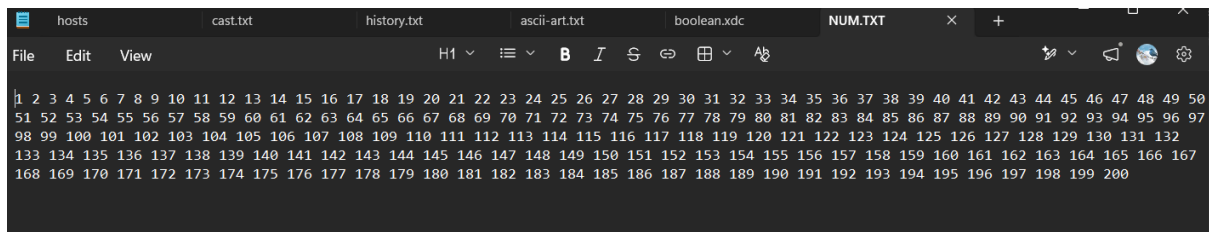
Code:-

```

1  //Assignment- 9
2  //Question -1
3  #include <iostream>
4  #include <fstream>
5  using namespace std;
6
7  int main() {
8      ofstream file("NUM.TXT");
9
10     for(int i = 1; i <= 200; i++) {
11         file << i << " ";
12     }
13
14     file.close();
15     cout << "Numbers written successfully!";
16     return 0;
17 }

```

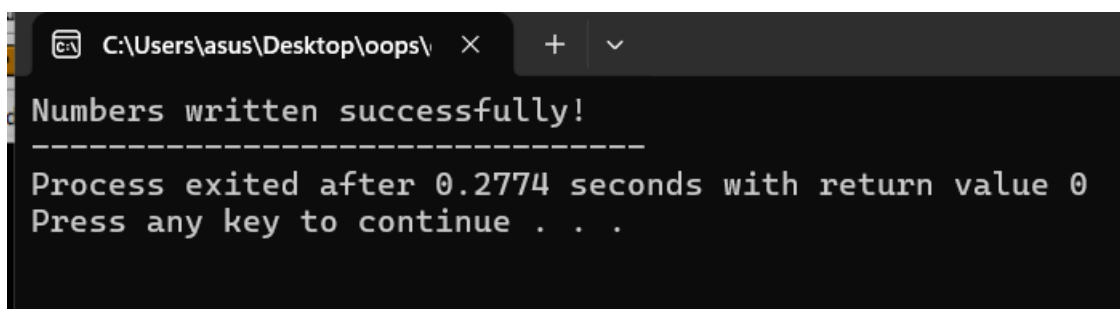
Output:-



```

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50
51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97
98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132
133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167
168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200

```



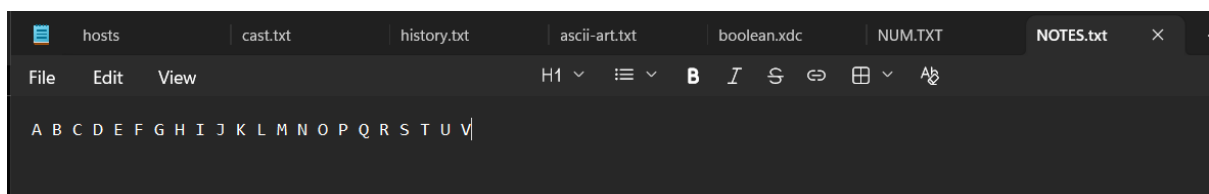
```

Numbers written successfully!
-----
Process exited after 0.2774 seconds with return value 0
Press any key to continue . . .

```

2. Write a user-defined function in C++ to read the content from a text file NOTES.TXT, count and display the number of alphabets present in it.

Code:-



```

A B C D E F G H I J K L M N O P Q R S T U V |

```

```

ques2.cpp [*] ques1.cpp
1 //Assignment- 9
2 //Question -4
3 #include <iostream>
4 #include <fstream>
5 #include <cctype>
6 using namespace std;
7
8 void countAlphabets() {
9     ifstream file("NOTES.TXT");
10    char ch;
11    int count = 0;
12
13    while(file.get(ch)) {
14        if(isalpha(ch)) {
15            count++;
16        }
17    }
18
19    cout << "Total alphabets: " << count;
20    file.close();
21 }
22
23 int main() {
24    countAlphabets();
25    return 0;
26 }

```

Output:-

```

C:\Users\asus\Desktop\oops\ x + v
Total alphabets: 22
-----
Process exited after 0.2115 seconds with return value 0
Press any key to continue . . .

```

3. Write a program to perform read/write binary I/O operation on a file (i.e. write the object of the class to a file)

Code:-

```

hosts cast.txt history.txt ascii-art.txt boolean.xdc NUM.TXT NOTES.txt source.txt dest.txt studen x +
File Edit View
e 情j @ 葵j

```

```

ques3.cpp
1  //Assignment- 9
2  //Question -4
3  #include <iostream>
4  #include <fstream>
5  using namespace std;
6  class Student {
7      int roll;
8      char name[20];
9  public:
10     void input() {
11         cout << "Enter Roll and Name: ";
12         cin >> roll >> name;
13     }
14     void display() {
15         cout << "Roll: " << roll << " Name: " << name << endl;
16     }
17 };
18 int main() {
19     Student s;
20     ofstream fout("student.dat", ios::binary);
21     s.input();
22     fout.write((char*)&s, sizeof(s));
23     fout.close();
24     ifstream fin("student.dat", ios::binary);
25     Student s2;
26     fin.read((char*)&s2, sizeof(s2));
27     cout << "\nData read from file:\n";
28     s2.display();
29     fin.close();
30 }

```

```

C:\Users\asus\Desktop\oops\  ×  +  ▾
Enter Roll and Name: 101 Raj

Data read from file:
Roll: 101 Name: Raj

-----
Process exited after 4.305 seconds with return value 0
Press any key to continue . . . |

```

4. Write a program to copy the contents of one file to another.

Code:-

```

hosts  cast.txt  history.txt  ascii-art.txt  boolean.xdc  NUM.TXT  NOTES.txt  source.txt  ×  +
File  Edit  View  H1  ▾  ≡  ▾  B  I  S  ⇄  ▢  ▾  A  ▾  ↗  ▾
"Hello India"

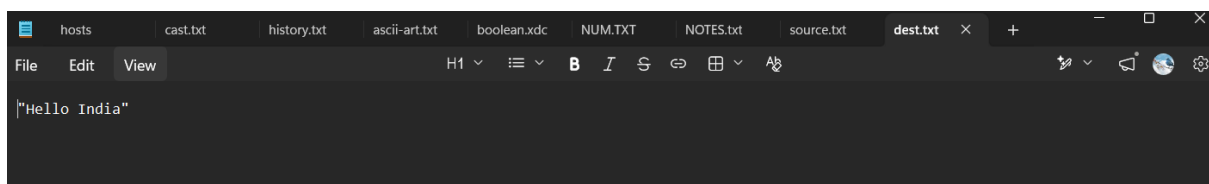
```

```

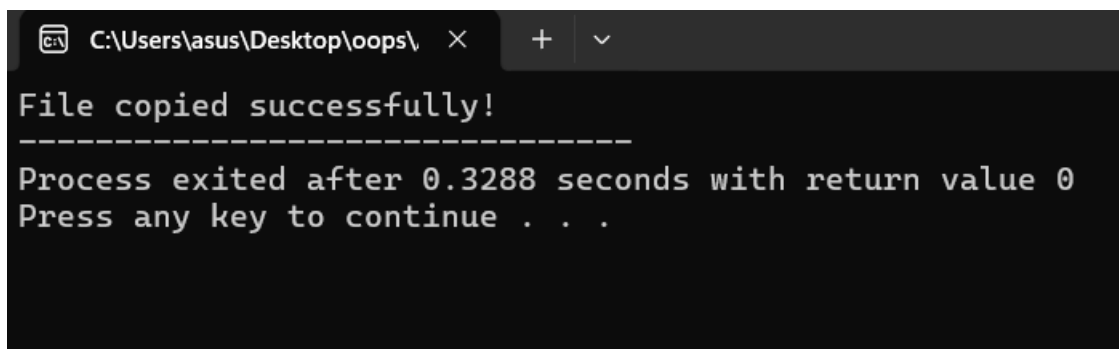
ques3.cpp  ques4.cpp
1  //Assignment- 9
2  //Question -4
3  #include <iostream>
4  #include <fstream>
5  using namespace std;
6  int main() {
7      ifstream source("source.txt");
8      ofstream dest("dest.txt");
9      char ch;
10     while(source.get(ch)) {
11         dest.put(ch);
12     }
13     cout << "File copied successfully!";
14     source.close();
15     dest.close();
16 }

```

Output:-



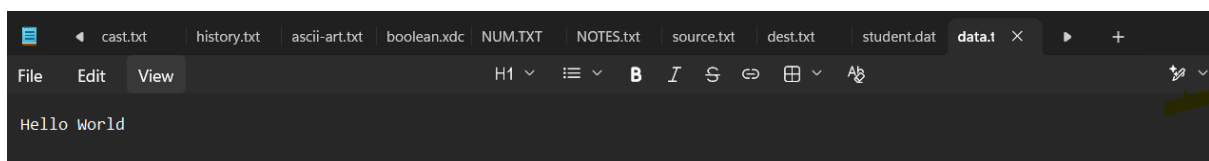
A screenshot of a text editor window with multiple tabs. The active tab is 'dest.txt', which contains the text "Hello India".



A screenshot of a command prompt window. It displays the output "File copied successfully!" followed by a separator line. Below the separator, it shows "Process exited after 0.3288 seconds with return value 0" and "Press any key to continue . . .".

5. Write a program to implement I/O operations on characters. I/O operations include inputting a string, calculating the length of the string, storing the string in a file and fetching the stored characters from it.

Code:-



A screenshot of a text editor window with multiple tabs. The active tab is 'data.txt', which contains the text "Hello World".

```

ques3.cpp  ques4.cpp  [*] Untitled2  ques5.cpp
1  //Assignment-9
2  //Question-5
3  #include <iostream>
4  #include <fstream>
5  #include <string>
6  using namespace std;
7
8  int main() {
9      string str;
10
11      cout << "Enter a string: ";
12      getline(cin, str);
13
14      cout << "Length: " << str.length() << endl;
15
16      ofstream file("data.txt");
17      file << str;
18      file.close();
19
20      ifstream file2("data.txt");
21      string output;
22      getline(file2, output);
23
24      cout << "Read from file: " << output;
25
26      file2.close();
27      return 0;
28 }

```

Output:-

```

C:\Users\asus\Desktop\oops\  X  +  v
Enter a string: Hello World
Length: 11
Read from file: Hello World
-----
Process exited after 5.961 seconds with return value 0
Press any key to continue . . .

```

6. (a) Create a file A-Z and read 10th character using seekg().

Code:-

```

history.txt  ascii-art.txt  boolean.xdc  NUM.TXT  NOTES.txt  source.txt  dest.txt  student.dat  data.txt  AZ.txt  X  ►
File  Edit  View  H1  v  ≡  v  B  I  S  ⇌  ▢  v  Aa
ABCDEFGHIJKLMNOPQRSTUVWXYZ

```

```

ques3.cpp  ques4.cpp  ques6a.cpp  ques5.cpp
1  //Assignment- 9
2  //Question -6(a)
3  #include <fstream>
4  #include <iostream>
5  using namespace std;
6
7  int main() {
8      ofstream file("AZ.txt");
9      for(char ch = 'A'; ch <= 'Z'; ch++)
10         file << ch;
11         file.close();
12
13         ifstream in("AZ.txt");
14         in.seekg(9); // 10th char (index starts from 0)
15
16         char ch;
17         in.get(ch);
18         cout << "10th character: " << ch;
19
20         in.close();
21     }

```

Output:-

```

C:\Users\asus\Desktop\oops\  x  +  v
10th character: J
-----
Process exited after 0.3328 seconds with return value 0
Press any key to continue . . .

```

6(b) Overwrite 5th number using seekp().

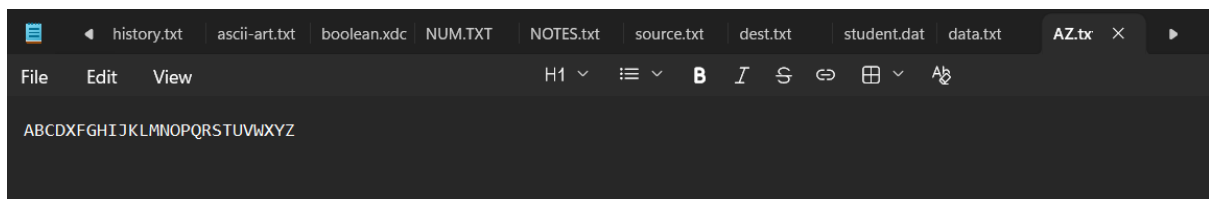
Code:-

```

ques3.cpp  ques4.cpp  [*] ques6a.cpp  ques5.cpp  [*] ques6b.cpp
1  //Assignment- 9
2  //Question -6(b)
3  #include <fstream>
4  #include <iostream>
5  using namespace std;
6
7  int main() {
8      fstream file("AZ.txt", ios::in | ios::out);
9      file.seekp(4); // 5th position
10     file.put('X');
11
12     file.close();
13 }

```

Output:-



6(c) Find file size using tellg().

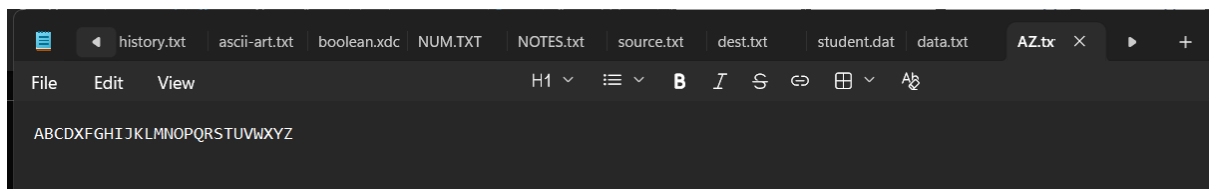
Code:-

```

ques3.cpp  ques4.cpp  [*] ques6a.cpp  ques5.cpp  [*] ques6b.cpp  ques6c.cpp
1  //Assignment- 9
2  //Question -6(c)
3  #include <fstream>
4  #include <iostream>
5  using namespace std;
6  int main() {
7      ifstream file("AZ.txt");
8
9      file.seekg(0, ios::end); // go to end
10     cout << "File size: " << file.tellg() << " bytes";
11
12     file.close();
13 }

```

Output:-



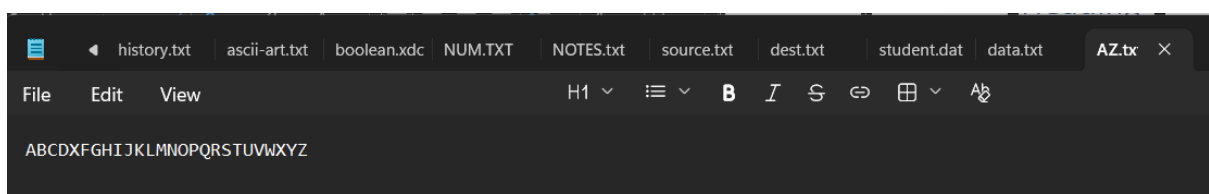
```

C:\Users\asus\Desktop\oops\  X  +  v
File size: 26 bytes
-----
Process exited after 0.3208 seconds with return value 0
Press any key to continue . . .

```

6(d) Read last character using seekg().

Code:-



```

ques3.cpp  ques4.cpp  [*] ques6a.cpp  ques5.cpp  [*] ques6b.cpp  ques6c.cpp  ques6d.c
1  //Assignment- 9
2  //Question -6(d)
3  #include <fstream>
4  #include <iostream>
5  using namespace std;
6
7  int main() {
8      ifstream file("AZ.txt");
9
10     file.seekg(-1, ios::end);
11     char ch;
12     file.get(ch);
13
14     cout << "Last character: " << ch;
15
16     file.close();
17 }

```

Output:-

```

C:\Users\asus\Desktop\oops\  X  +  v
Last character: Z
-----
Process exited after 0.3019 seconds with return value 0
Press any key to continue . . .

```

6(e) Create a file data.txt with multiple lines. Read the file and:

- i. Move the pointer to the 10th byte using seekg()
- ii. Display the current position using tellg().
- iii. Read and display the remaining content.

Code:-

```

ques3.cpp  ques4.cpp  [*] ques6a.cpp  ques5.cpp  [*] ques6b.cpp  ques6c.cpp  ques6d.cpp  ques6e.cpp
1  //Assignment- 9
2  //Question -6(e)
3  #include <fstream>
4  #include <iostream>
5  using namespace std;
6  int main() {
7      ofstream out("data1.txt");
8      out << "Hello this is a sample file.\nIt has multiple lines.\nEnjoy learning!";
9      out.close();
10
11     ifstream in("data1.txt");
12
13     in.seekg(10);
14     cout << "Position: " << in.tellg() << endl;
15
16     char ch;
17     while(in.get(ch)) {
18         cout << ch;
19     }
20
21     in.close();
22 }

```


Output:-

The top image shows a text editor window with the following text:

```
Hello this is a sample file.
It has multiple lines.
Enjoy learning!
```

The bottom image shows a terminal window with the following output:

```
Position: 10
 is a sample file.
It has multiple lines.
Enjoy learning!
-----
Process exited after 0.3011 seconds with return value 0
Press any key to continue . . .
```

7. Write a program to:

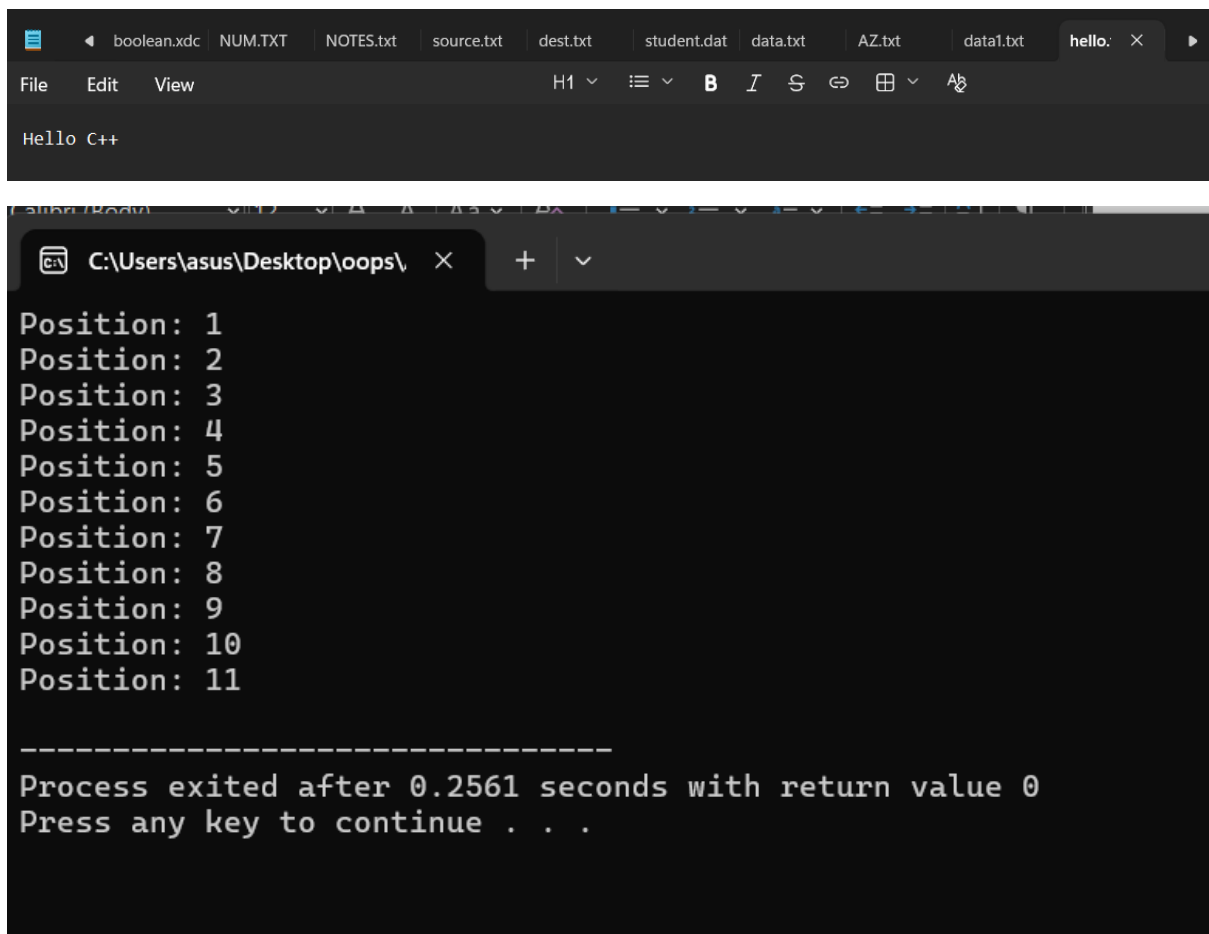
- Create a file with text "HelloWorld"
- After writing each character, display the current position of the put pointer using `tellp()`.
- Replace "World" with "C++" using random access

Code:-

```
ques3.cpp ques4.cpp [*] ques6a.cpp ques5.cpp [*] ques6b.cpp ques6c.cpp ques6d.cpp ques6e.cpp ques7.cpp
1 //Assignment- 9
2 //Question -7
3 #include <iostream>
4 #include <fstream>
5 #include <string>
6 using namespace std;
7
8 int main() {
9     fstream file("hello.txt", ios::out | ios::in | ios::trunc);
10
11     if (!file) {
12         cout << "File error!";
13         return 0;
14     }
15
16     string text = "Hello World";
17
18     // Write characters and show position
19     for(int i = 0; i < text.length(); i++) {
20         char ch = text[i];
21         file.put(ch);
22         cout << "Position: " << file.tellp() << endl;
23     }
24
25     // Replace "World" with "C++"
26     file.seekp(6); // 'W' position
27     file << "C++ "; // overwrite
28
29     file.close();
30
31     return 0;
32 }
```

Akshita_1024150293

Output:-



The image shows a C++ IDE window with multiple tabs. The active tab is 'hello:'. The code in the editor is 'Hello C++'. Below the editor is the output window, which displays the following text:

```
Position: 1
Position: 2
Position: 3
Position: 4
Position: 5
Position: 6
Position: 7
Position: 8
Position: 9
Position: 10
Position: 11

-----
Process exited after 0.2561 seconds with return value 0
Press any key to continue . . .
```